MILITARY SPECIFICATION

PINION AND BUSHING ASSEMBLY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of pinion and bushing assembly used on fuze for M422 Projectile.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipments of Inert Ammunition Components

DRAWINGS

ORDNANCE CORPS

8593086 - Pinion and Bushing Assembly
8593087 - Pinion
8593117 - Bushing, Pinion
8808362 - Packing and Marking for Storage and Shipments
8883538 - Bag, Packing, for Repair Parts
ET-8593086 - Equipment Tabulation

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The pinion and bushing assembly shall comply with all requirements specified on the drawing(dwg) 8593086, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Workmanship.

3.3.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.3.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.3.3 Burrs. Parts shall be free from burrs.

3.3.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.3.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Quality conformance inspection.

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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table I and Table II), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table I
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table II
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131			

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Pin, bushing (see dwg 8593118) covering a detail of dwg 8593086.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Outside diameter (.0630)	Gage
102.	Length (.140)	Gage
103.	Material (see 3.1)	Visual
104.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8593118	Visual
Minor:		
201.	Radius or chamfer present	Visual
202.	Workmanship (see 3.3)	Visual/Manual

4.3.2.1.2 Bushing, pinion (see dwg 8593117) covering a detail of dwg 8593086.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Inside diameter (.2505)	Gage
102.	Concentricity of inside diameter	Gage
103.	Diameter of 3 holes (.077)	Gage

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104. Location of 3 holes.....Gage
105. Small outside diameter (.455).....Gage
106. Diameter of pin hole (.0620).....Gage
107. Location of pin hole (.085).....Gage
108. Material (see 3.1).....Visual

Minor:

201. Diameter of flange (.750).....Gage
202. Overall length (.31).....Gage
203. Length of small outside diameter (.171).....Gage
204. Presence of 3 countersinks.....Visual
205. Proper surface finish.....Visual
206. Chamfer or radius present.....Visual
207. Workmanship (see 3.3).....Visual/Manual

4.3.2.1.3 Pinion (see dwg 8593087) covering a detail of dwg 8593086.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Central outside diameter (.160).....Gage
102. Diameter right and left of central outside diameter (.2496).....Gage
103. Outside diameter right of teeth (.320).....Gage
104. Length of central outside diameter.....Gage
105. Length from left end to shoulder of outside diameter adjacent to teeth (.143).....Gage
106. Length from shoulder of outside diameter adjacent to teeth to right outside diameter (.119).....Gage
107. Length of outside diameter adjacent to teeth (.010).....Gage
108. Distance across flats of hexagonal hole (.125).....Gage
109. Depth of hexagonal hole to end of tapered part, max (.150).....Gage
110. Depth of hexagonal hole to end of flats, min (.109).....Gage
111. Overall length, max (.453).....Gage
112. Length from shoulder of outside diameter adjacent to teeth to right surface (.31).....Gage
113. Gear characteristics (functional).....Gage
114. Material (see 3.1).....Visual
115. Objective evidence (see 6.2) for protective finish in accordance with dwg 8593087.....Visual

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Minor:

201. Proper surface finish.....Visual
202. Workmanship (see 3.3).....Visual/Manual

4.3.2.1.4 Pinion and bushing assembly (see dwg 8593086)

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Pinion rotates freely (see 4.3.3.1).....Manual

Minor:

201. Workmanship (see 3.3).....Visual/Manual

4.3.2.1.5 Preparation for delivery (see 5.1) .

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Preservation, packaging, packing and marking shall be in accordance with MIL-STD-1169.....Visual

Minor: None defined.

4.3.2.1.6 Bag, packing, for repair parts (see dwg 8883538 covering dwg part No. 8883538-2) and 5.2.

Categories	Characteristics	Method of Inspection

Critical: None defined.

Major:

101. Material (see 3.1).....Visual
102. Length before packing, min.....*Measure
103. Width before packing, min.....*Measure
104. Properly packed.....Visual/Manual
105. Proper heat seal.....Visual/Manual
106. Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116.....Visual
107. No tear, crack or hole in bag.....Visual

*Measure with suitable scale.

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Minor:

201. Proper marking.....Visual
202. Workmanship (see 3.3).....Visual/Manual

4.3.2.1.7 Packing and marking for storage and shipments (see dwg 8808362) and 5.2.

Categories	Characteristics	Method of Inspection

Critical: None defined

Major:

101. Gross load weight, max.....*Measure
102. Contents tight in package.....Manual
103. Correctly packed.....Visual/Manual
104. Material (see 3.1).....Visual

*Determine weight with suitable scale.

Minor:

201. Proper marking.....Visual
202. Workmanship (see 3.3).....Visual/Manual

4.3.3 Testing. This test shall be performed.

Test	Test Classification
a. Rotation of pinion	Major

This test is not destructive and parts so tested may be returned to the lot.

4.3.3.1 Rotation of pinion (see dwg 8593086). Each pinion and bushing assembly shall be inspected for freedom of rotation as specified in 4.4.1. If any assembly fails to comply with dwg requirements, the assembly shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-8593086.. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

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a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Rotation of pinion (see 4.3.3.1). Each pinion and bushing assembly shall be tested for freedom of rotation by grasping the large flange in one hand and rotating the pinion with the other hand.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The assembly shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The assembly shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The assembly shall be packaged in accordance with dwgs 8883538 and 8808362 using the specified moisture barrier.

5.2.1.2 Level C. The assembly shall be packaged in accordance with dwg 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing

5.2.2.1 Level A and C. The assembly shall be packed in accordance with dwgs 8883538 and 8808362.

5.2.3 Marking Marking shall be as specified in accordance with dwgs 8883538 and 8808362 as applicable.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements

a. Title, number, and date of this specification.
b. Production or Repair Parts.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-N-I.

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